



EXAMINATION

MJ2530/240SELKTH9 Thermal Conversion and Industrial Energy Systems Analysis

Part A: Home Examination

May 26th 2025

09:00-17:00

Part B: Oral Examination

May 30th, June 2nd & June 16th 2025

10 minutes per student according to individual schedule

Aids allowed for Part A: Home Exam

- **No communication with anyone!**
- Literature, including on-line resources relevant to the course
- Lecture notes, hand-outs, and solved homework assignments from the course

Submission of Part A: Take Home Exam

- One written document (word/pdf) describing your solution to the problem, including the steps of the analysis and key equations, assumptions made and a short motivation of these, results (graphs and tables), and final analysis and discussion. The majority of your grade will be given based on the clarity of this document.
- The “source code” for the computational part, regardless of choice of programme (Excel worksheet, EES or MATLAB code, etc.). **Source code is NOT a valid document to show your solution.**

- Material has to be submitted through a special assignment in “Assignments” in the Canvas course event MJ2530 VT25 Thermal Conversion and Industrial Energy Systems Analysis.
- **Submission Deadline: 17:00 on May 26th 2025 – No exceptions!**

Aids allowed for Part B: Oral Examination

- Only the student’s own copy of the solution submitted via Canvas for Part A.

Grading

Maximum Score: 100 points

Grade	
A	100 – 90
B	90 – 76
C	76 – 62
D	62 – 55
E	55 – 50
Fx	50 – 49
F (fail)	< 49

The results will be posted no later than on June 27th 2025.

Good luck!

Physical Model

Simple gas turbine fed by CH₄ fuel:

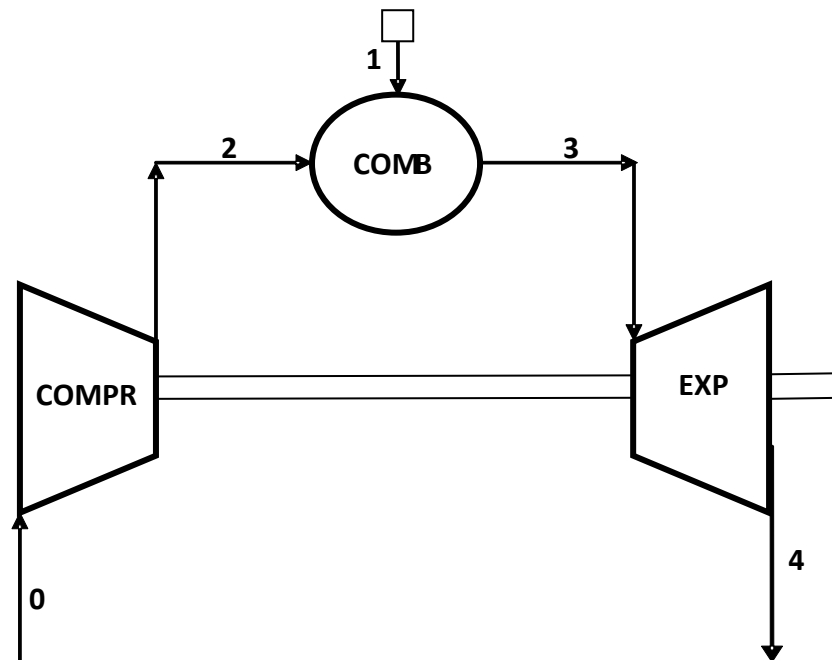


Figure 1. Physical Model of the power plant

FLOW	TYPE
0	External air
1	Fuel (CH ₄)
2	Compressed air
3	Gases (mixture)
4	Gases (mixture)

Table 1. Flows

Some technical indications

PLANT PRODUCT: $W_{el,net} = 50$ MW

EXTERNAL AIR: $T_0 = 25^\circ\text{C}$ $p_0 = 1$ bar

Cost functions

Air compressor

$$Z_1 = \frac{c_{11} \cdot G_a}{c_{12} - \eta_c} \cdot \beta \cdot \ln \beta$$

Combustor

$$Z_2 = \frac{c_{21} \cdot G_a}{c_{22} - r_b} \cdot [1 + \exp(c_{23} \cdot T_3 - c_{24})]$$

Gas Turbine (expander)

$$Z_3 = \frac{c_{31} \cdot G_g}{c_{32} - \eta_t} \cdot \ln r_t \cdot [1 + \exp(c_{33} \cdot T_3 - c_{34})]$$

The economic parameters used in the investment cost functions are reported in Table 2:

$c_{11} = 39.5 \text{ \$ / (kg/s)}$	$c_{24} = 26.4$
$c_{12} = 0.9$	$c_{31} = 266.3 \text{ \$ / (kg/s)}$
$c_{21} = 25.6 \text{ \$ / (kg/s)}$	$c_{32} = 0.92$
$c_{22} = 0.995$	$c_{33} = 0.036 \text{ K}^{-1}$
$c_{23} = 0.018 \text{ K}^{-1}$	$c_{34} = 54.4$

Table 2. Economic parameters in the investment cost functions of the plant components

Economic assumptions

To evaluate the economic cost (of components and fuel), the following terms have to be fixed:

- Capital Recovery Factor (CRF) See Table 3 (Individual case data)
- Maintenance Factor: $\phi = 1.10$
- Number of operation hours per year (N) See Table 3 (Individual case data)
- Fuel (CH₄) cost (c_f) See Table 3 (Individual case data)

PERFORM:

Thermoeconomic optimization of the plant:

1. Perform a thermo-economic optimization of the plant, in nominal conditions
2. Make a sensitivity analysis of the optimal result when varying the cost of the fuel (CH₄) from -100% to +100% of the value indicated in the initial data

Environomic optimization of the plant:

3. Consider the CO and the NO_x as the contaminant emitted by the plant: find in the literature a suitable value of specific CO and NO_x emissions from natural gas fed gas turbines, then add it in the environomic objective function, considering the pollution penalty factor f_p and the cost of emission here reported:
pollution penalty factor $f_{p,CO}$ equal to 7.0; environomic cost of CO emission of 3 \$/kg;
pollution penalty factor $f_{p,NOx}$ equal to 1.1; environomic cost of NO_x emission of 5 \$/kg;
Find the optimal result, considering the cost of CH₄ in the nominal conditions.
4. Make a sensitivity analysis of the optimum result towards the environomic cost of kg emitted of CO and NO_x, varying them from +20% to +100%, still considering the cost of CH₄ in the nominal conditions

Comparison

5. Make a cross comparison between the effects of the cost of CH₄ (from point 2 above) and the imposition of a tax over the CO and NO_x emissions (from point 4 above): which are your evaluations comparing the two situations?

Write a short report of your results (possibly, making use of graphical representations of the results) and be able to discuss the results in the oral exam.

Individual case data

The individual case data are shown in Table 3.

Table 3. Individual case data

Case no	Fuel cost (c_f) [10 ⁻⁶ USD/kJ]	(methane)	Number of operation hours per year (N)	Capital Recovery Factor (CRF)
1	7		6500	0.20
2	7		6500	0.16
3	7		7000	0.16
4	7		7000	0.18
5	7		7000	0.20
6	7		7500	0.16
7	7		7500	0.18
8	7		7500	0.20
9	7		8000	0.16
10	7		8000	0.18
11	7		8000	0.20
12	8		8000	0.16
13	8		8000	0.18
14	8		8000	0.20
15	8		7500	0.16
16	8		7500	0.18
17	8		7500	0.20
18	8		7000	0.16
19	8		7000	0.18
20	8		7000	0.20
21	8		6500	0.16
22	9		7000	0.16
23	9		7000	0.18
24	9		7000	0.20
25	9		7500	0.16
26	9		7500	0.18
27	9		7500	0.20
28	9		8000	0.16
29	9		8000	0.18
30	9		8000	0.20
31	9		6500	0.16
32	9		6500	0.18
33	9		6500	0.20
34	9		8500	0.16
35	9		8500	0.18
36	9		8500	0.20
37	10		6500	0.16
38	10		6500	0.18
39	10		6500	0.20
40	10		7000	0.16

41	10	7000	0.18
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